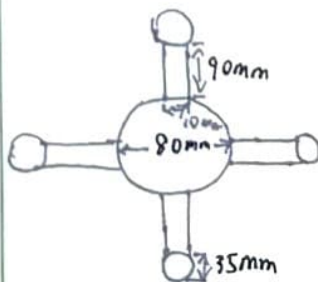


Assump: Set Battery Capacity less 2000 mAh. 12S with 2 of 6S Battery series connect
I choose C=1800 mAh Battery this needs 80 mm Diameter tube 5mm thickness
set $C_d = 0.3$



$$A = \frac{\pi}{4}(80)^2 + 4 \cdot 90 \cdot 10 + \frac{\pi}{4}(35)^2 \cdot 4 ; \text{Density: } \rho = 1,225 \text{ kg/m}^3$$

$$= 0.0125 \text{ m}^2$$

$$\text{Velocity: } V = 600 \text{ km/h} \approx 166.67 \text{ m/s}$$

$$\approx 170 \text{ m/s}$$

$$C_d \cdot A = 0.00375$$

$$\text{Drag: } D = \frac{1}{2} \rho \cdot V^2 \cdot C_d \cdot A = \frac{1}{2} \cdot 1,225 \cdot 170^2 \cdot 0.00375$$

$$= 66.38 \text{ N}$$

$$P_{\text{drag}} = D \cdot V = 66.38 \cdot 170 = 11.29 \text{ kW}$$

Assume:

$$\text{efficiency: } \eta = 0.55 \rightarrow P_{\text{motor}} = \frac{P_{\text{drag}}}{\eta} = 20.53 \text{ kW}$$

Find: the time for accelerate t

$$\frac{dE_k}{dt} = \frac{d(\frac{1}{2}mv^2)}{dt} = mV \frac{dv}{dt} = P_{\text{avail}} - P_{\text{drag}} = P_{\text{avail}} - \frac{1}{2} \rho \cdot V^3 \cdot C_d \cdot A$$

$$\frac{dv}{dt} = \frac{P_{\text{avail}}}{m} - \frac{1}{2} \rho C_d A V^2 \rightarrow \int dt = \int_0^{V_f} \frac{\frac{1}{2} m V dv}{P_{\text{avail}} - \frac{1}{2} \rho C_d A V^3}$$

time for accelerating from 0-170 m/s at

$$t = \int_0^{170} \frac{3 \cdot V}{\frac{8000}{24000} - \frac{1}{2} \cdot 1,225 \cdot 0.00375 V^3} dv = 3.41 \text{ s}$$

Assume & known condition

$$m = 3 \text{ kg}$$

$$\rho = 1,225 \text{ kg/m}^3$$

$$C_d \cdot A = 0.00375$$

$$P_{\text{avail}} = 24 \text{ kW assumed } 24 \text{ kW}$$

$$V_f = 170 \text{ m/s}$$

$$P_{\text{avail}} = 6 \text{ kW } 4 \cdot 6 = 24 \text{ kW}$$

Energy for accelerating

$$E = t \cdot P_{\text{avail}} = 25 \text{ kW} \cdot \frac{3.41 \text{ s}}{3600 \text{ s/h}} = 23.68 \text{ Wh}$$

$$\text{average voltage for 12S } V = 48 \text{ V} \rightarrow \frac{23.68 \text{ Wh}}{48 \text{ V}} = 493.35 \text{ mAh} \approx 500 \text{ mAh}$$

VS Battery 1800 mAh

But we need the motor that can output 6kW power, or the time will increase
is $2t \rightarrow 1000 \text{ mAh}$ to accelerate

Pusher V.S. Puller

$$\text{Drag: } D_c = \frac{1}{2} \rho V^2 S \cdot C_{D,0}$$

- Parameter just guess (no logic derivation)

$$\text{Power balance: } P_{\text{avail}} = P_{\text{shaft}} \cdot \eta = P \cdot V$$

Assume: $\rho_{\text{air}} = 1.225 \text{ kg/m}^3$, reference Area: $S_{\text{ref}} = [0.02, 0.05] \text{ m}^2$ I will assume 0.03 m^2

Diameter Prop: $D_{\text{prop}} = 0.185 \text{ m}$, Propeller efficiency: $\eta_0 = 0.88$, Parasite drag coefficient $C_{D,0} = 0.035$

Push: efficiency loss: $\phi = 0.07$ [0.03-0.07] usually range

pull: Area affected by prop: $S \cdot C_D = 0.01 \text{ m}^2$

$P = 3 \text{ kW}$, if $V = 100 \text{ m/s}$ or 150 m/s
(360 km/h) (540 km/h)

~~Puller:~~

$$D_c = \frac{1}{2} \rho V^2 \cdot S \cdot C_{D,0} = \frac{1}{2} (1.225) (100)^2 \cdot 0.03 \cdot 0.035 = 6.43 \text{ N}$$

Puller

$$T = \frac{P_c}{\rho A_{\text{disk}} V^2}, \quad V_s = V \sqrt{1 + 2T}, \quad \Delta D_{\text{puller}} = \frac{1}{2} \rho (V_s^2 - V^2) \quad \left| \begin{array}{l} A_{\text{disk}} = \frac{\pi}{4} \cdot (0.1778)^2 = 0.0099 \text{ m}^2 \\ V = 100 \text{ m/s} \end{array} \right.$$

$$= \frac{6.43}{1.225 (0.0099) (100)^2} = 100.529 \text{ m/s} \quad = 0.6497 \quad \left| \begin{array}{l} \cdot S \cdot C_{D,0} \\ = 0.6497 \end{array} \right.$$

$$= 0.0053$$

$$\eta_{\text{puller}} = 0.88, \quad V_{\text{max}} = \frac{P \cdot \eta_{\text{puller}}}{D} = P \cdot \frac{0.88}{7.0797} = 0.124 P = 372$$

Pusher

$$\eta_{\text{pusher}} = \eta_0 \cdot \eta_{\text{ideal}} (1 - \phi)$$

$$= 0.88 (1 - 0.07)$$

$$= 0.818$$

$$V_{\text{max}} = \frac{P \cdot \eta_{\text{pusher}}}{D} = P \cdot \frac{0.818}{6.43} = 0.127 P = 381$$

- adjustable
- required design constants

1) target: $V_{max} = 170 \text{ m/s} \approx 612 \text{ km/h}$; time $t \leq 60 \text{ s (1 min)}$

limitation of Power system: Battery ~~12S~~ - 12S, Capacity $\leq 3000 \text{ mAh}$

Propeller - 7x15 (maybe 6x16)

ESC - can be cooling by water

2) Condition & Assumption

Density $\rho = 1.225 \text{ kg/m}^3$

Pitch 15 inch $\approx 0.381 \text{ m/round}$

Area $A = 0.0125 \text{ m}^2 = C_d \cdot A = 0.00125 \text{ m}^2$

$C_d = 0.1$

efficiency: $\eta = 0.55$

mass $m \leq 3 \text{ kg}$

Voltage $V_{bus} = 42 \text{ V (3.5 avg)}$

3) Calculation of Drag, Power

Drag: $D = \frac{1}{2} \rho V^2 C_d A = \frac{1}{2} (1.225) (170)^2 (0.00125) = 22.127 \text{ N}$

Power Drag: $P_{drag} = D \cdot V = 22.127 \cdot 170 = 3.762 \text{ kW}$

Motor Power: $P_{motor} = \frac{P_{drag}}{\eta} = \frac{3.762}{0.55} = 6.839 \text{ kW}$

- Motor Power each one needs 1.71 kW per motor

- Current $I = \frac{P}{V_{bus}} = \frac{1.71 \text{ kW}}{42 \text{ V}} = 40.71 \text{ A}$ per motor

- n (round per min) $n = \frac{V}{\text{pitch}} = \frac{170 \text{ m/s}}{0.381 \text{ m/round}} \cdot \frac{60 \text{ s}}{\text{min}} = \frac{26771.7}{0.8} \text{ rpm}$

- choosing 'kv' for motor

experience equation: $n \approx 0.8 \cdot 'kv' \cdot V_{bus} \rightarrow 'kv' = \frac{26772}{0.8 (42)} = 797 \text{ KV}$

range [700, 800, 900]

4) Battery (Capacity, 'C' value, 125)

- time for accelerate

$$\frac{dE_k}{dt} = P_{avail} - P_{drag} = P_{avail} - \frac{1}{2} \rho v^3 C_d A$$

$$t = \int_0^{v_t} \frac{m v(t) dv}{P_{avail} - \frac{1}{2} \rho C_d A v^3} = \int_0^{170} \frac{3 \cdot v}{11.16000 - 6839.7162} dv = 8.6s$$

Base on AMAX 2826 [3.5 - 4.6] take 4kW

- Energy for accelerating

$$E = t \cdot P_{avail} = \overset{(8.6)}{8.6} \cdot 95.16 \text{ kW} = 40 \text{ Wh} \rightarrow \frac{40 \text{ Wh}}{42 \text{ V}} \approx 952.4 \text{ mAh}$$

~~→~~

- C value